PORTFOLIO BLOCK 2 • TECHNICAL ARCHITECTURE REPORT

Enterprise Financial Formula Auditor & Dynamic Error Repair Engine

Cloud Architecture & Automated Data Hygiene Pipeline (Google Sheets Version)

1. Executive Summary

Financial reporting pipelines frequently ingest transactional feeds from disparate ERP systems containing unvalidated string formats, blank vendor references, and execution faults. This document outlines the architectural specifications of an enterprise-grade cloud auditing and dynamic error-repair workbook built in Google Sheets. The solution establishes automated sanitization layers, absolute variance reconciliation against subledger control accounts within a strict **$0.04 tolerance threshold**, and real-time executive visibility.

> **Core Operational Objectives**
> Autonomous trapping of calculation errors (`#N/A` , `#DIV/0!` , `#VALUE!`), programmatic currency string parsing, and automated zero-tolerance variance flagging across 100 historical general ledger transactions.

2. Multi-Tab Cloud Pipeline Architecture

The workbook is structured into 4 sequential operational tabs designed for immutable data flow from raw ingestion to executive presentation:

Tab Name	Architectural Role	Primary Function & Logic
`General_Ledger_Data`	Ingestion Layer	Stores 100 raw ERP transaction records containing deliberate corporate anomalies and errors.
`Formula_Repair_Engine`	Sanitization Engine	Applies cloud-native error-trapping and string parsing functions to coerce bad data.
`Reconciliation_Matrix`	Variance Engine	Compares repaired costs against subledger balances within a $0.04 threshold.
`Executive_Dashboard`	Command Console	Aggregates KPIs, audit volumes, and system health stability scores in real time.

3. Cloud-Native Formula Repair Specifications

To operate seamlessly within Google Sheets, the sanitization layer (Tab 2) utilizes robust regex and logical wrapper functions to cleanse raw transactional variables before financial aggregation:

Target Field	Google Sheets Repair Formula	Operational Effect
`Vendor_ID` (Col D)	`=IF(OR(ISBLANK(...), ...="#N/A"), "VEND-UNKNOWN", ...)`	Traps missing vendor references and replaces `#N/A` lookups.
`Clean_Quantity` (Col E)	`=IFERROR(VALUE(REGEXREPLACE(TO_TEXT(...), "[^0-9.]", "")), 0)`	Strips text corruptions (e.g. `INVALID_QTY`) into numeric values.
`Clean_Unit_Price` (Col F)	`=IFERROR(VALUE(REGEXREPLACE(TO_TEXT(...), "[^0-9.]", "")), 0)`	Strips currency symbols (`$`) and spaces into valid floats.
`Extended_Cost` (Col G)	`=IF(OR(E2=0, F2=0), 0, E2*F2)`	Prevents `#DIV/0!` calculation crashes on zero-value units.
`Reconciliation` (Col F3)	`=IF(D2<=0.04, "MATCHED", "VARIANCE DETECTED")`	Enforces the strict $0.04 variance control limit.

4. Executive Dashboard & System Health Metrics

Tab 4 synthesizes operational integrity across the audit pipeline, presenting key performance indicators and stability scores derived directly from downstream calculations:

TOTAL AUDITED	ERRORS REPAIRED	VARIANCES	SYSTEM HEALTH
100	**34**	**27**	**66.0%**

> **Apple Minimalist UI Standards**
> Designed with dark charcoal slate headers (`#222222`), pure white and ultra-light cool gray zebra striping (`#F9FBFD`), and breathable muted gridlines (`#E5E7EB`) to ensure maximum readability for executive presentations.